Just-in-Time Intervention for Augmenting AI Assistive Technology Personalization, Understanding, and Usability

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1 Introduction

1.1 Background

The use of mobile AI based assistive technology tools, particularly among the blind/visually impaired (BVI) community, is ubiquitous [22]. AI-powered mobile tools that help BVI individuals to navigate visual information—which range from text-recognition assistants [1, 2], to tools for generating image descriptions and exploring visual contexts [3, 18], to live video description [24], to navigational assistance [17]—have proliferated in both capability and quantity in recent years with the increasing integration of AI into daily technological life [10].

However, several barriers stand in the way of maximizing the utility of these tools. Firstly, a recent study documented a decrease in AI performance of up to 15% on visual data provided by blind individuals [22], making AI tools harder to use and trust. Common workarounds to this issue include drawing on sighted assistance— reducing the agency the tool nominally provides—and cross-checking across different tools—a manual, labor-intensive process that complicates time sensitive tasks [12]. Furthering the challenges associated with multi-app workflows, helpful features in one app—such as the camera aiming in SeeingAI's document mode [13]—cannot be readily ported to other apps that lack the feature but could benefit from its use—such as SeeingAI's image description—, in part due to major mobile operating system's policy of sandboxing apps, which forces them to run in environments isolated from each other [8, 27]. The effect of this restriction on inter-tool use is exacerbated by the often-lengthy gap between research and commercial practice: an approach to blind camera aiming assistance, for example, was documented in the academic literature as early as 2010 [31], yet, it did not surface in industry-designed, widely available mobile assistive technology until 2017 [13], and to this day, is not available in all tools for which it would be relevant. As of 2023, techniques for explainability and visual AI verification have emerged [14], but have yet to reach deployed systems. As AI continues to grow in prevalence, considering the stark performance gap for BVI users, these techniques cannot afford to wait years before becoming available.

Yet, as a product of the rapid proliferation of AI-based mobile assistive technologies, it is impractical to individually remediate all relevant apps with the features they would benefit from. Similarly, the majority of these apps are developed by corporations with greater resources and more dedicated personnel than an academic research team, so it is infeasible to develop a single app mobile AI-based assistive technology app that would compete with or replace the existing marketplace of tools.

1.2 Approach

With this problem space in mind, rather than developing findings from research into full applications, we aim to understand how such features could be accessed as a 'layer' to existing assistive applications [28]. In this way, not only can research be bridged with assistive technology workflows in practice, but users can have more abilities to customize and control their assistive technology experiences by configuring and accessing interventions as-needed.

1.2.1 Framework.

We present a framework and design space with example implementations for interventions for augmenting existing assistive applications for personalization, AI understanding, and usability. Inspired by work leveraging mobile automation tools such as Shortcuts on iOS to intervene on social media usage [11], we create a framework for use cases and surfacing methods to provide 'interventions'—additional features as add-ons to applications, providing relevant functionalities that may not otherwise be available. This framework, grounded in prior literature, lays out three central use categories: usability—interventions that promote using the tools as they already exist more effectively—, trust and confidence—for bolstering user trust in the tools they use—, and tool understanding—for supporting end-user understanding of why a tool might perform the way it performs. The first broad is driven by user experience research, and the latter two are rooted in AI explainability.

While we propose these categories separately, their impacts are complimentary. If a user can use a tool more effectively, they can be more confident in its outcomes. Likewise, if a user is more confident in a tool's outcomes, they can more easily understand the rationales behind its performance. Similarly, if a user understands

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why a tool performs how it performs, they are better equipped to use it effectively.

1.2.2 Interventions.

Beyond the framework itself, we present five sample interventions that encompass all three outlined utilities, as well as a variety of methods for surfacing and use.

In the category of effective use, we present camera-aiming, checking photo quality, and surfacing unknown-unknowns—which, in this setting, refers to relevant information that a user may not know is missing or compromised, leading to incomplete, ineffective, or even misleading situational understanding [7]. Surfacing camera aiming switches a user from their existing app to our layer app for real time camera aiming assistance and returns them to their original app once an acceptable framing has been reached. By contrast, photo quality checking can either be surfaced by a user in an external app, in which case their screen context is captured and processed externally while they remain in their original app, or can be surfaced inside the layer app through the submission of a pre-taken photo. With an additional intervention-triggering modality, surfacing unknown unknowns is called alongside other interventions for which it could be relevant—specifically those in the explainability rooted use categories, to be discussed later in this section.

In the category of confident usage, we present a cross-checking intervention. This allows users to verify the accuracy of some or all of a model's response, without the need to engage in the time-consuming process of manually checking with multiple tools. This intervention is triggered verbally by a user, and allows them to remain in their original app throughout the process, with processing by the layer app and ultimate delivery of a result confirming or denying the accuracy of the specified information occurring in the background. As mentioned, this intervention also triggers surfacing unknown-unknowns: providing additional information (if it exists) that may contextualize an incorrect result if the verification returns as incorrect, or providing additional context that may prove relevant even if the information a user wants to check is nominally accurate.

In the category of tool understanding, we present a debug intervention. This intervention allows a user to contest a model output. This intervention is called from a user's original app, and, like cross-checking, runs in the background. Depending on whether an app features on screen model output, the model response is either scraped from the onscreen-text or described by the user, and the user then specifies what they expected to find instead. A photo is then taken to capture the user's visual context, and they ultimately receive a brief explanation of why the model may have performed counter to their expectations. This intervention also triggers surfacing unknown-unknowns, which, if found, add additional context to the reasoning for a model's unexpected performance.

1.2.3 Evaluation.

We evaluate this work with a multi-phase user study, which includes several codesign sessions with two blind user-experts. Later, this codesign will be followed by a more traditional user study to examine user's preferences and for interventions and evaluate usability more broadly than was possible with codesign, and ultimately, an in-the-wild evaluation to examine the layer app's capacity to integrate into organic user workflows.

2 Methods

2.1 Codesign

This work is developed through codesign with two blind participants, both of whom work with mobile assistive technologies in their personal lives as well as professionally. Participant 1 is a tech trainer—teaching BVI individuals to use their phones and assistive technologies as a career—while Participant 2 works more closely with companies developing these tools in a consulting role.

Our codesign process is designed in three phases, with each participant having one session for each phase, for a total of six codesign sessions. Each session is an hour to an hour and a and a half in length. The first phase is designed to procure feedback on interventions as a concept and gauge user opinions on different ways these interventions could be surfaced. The second phase intends to evaluate early intervention prototypes, as well as iterate over surfacing modalities discussed as options in the first codesign phase to refine interaction workflows. The third phase builds upon where the second phase left off, testing refined versions of the interventions, exploring prototypes that were not yet testable as of phase two, and getting more precise feedback on the user experience using a variety of surfacing modalities.

By using codesign as a method, we intend to directly incorportate the feedback of BVI users throughout the entirety of the design process and mitigate the potential of overlooking user needs. Additionally, the career experience of our two participants lends valuable insight to potential perspectives from users beyond just the two participants: Participant 1 brings a perspective informed by the needs of his tech-training clients, who may be reluctant or unable to dive into complicated workflows, while Participant 2 brings a corporate perspective, as well as one of a more tech-forward user interested in fully optimizing his assistive technology workflow. Through this range of experiences and perspectives, we hope to gain insights from codesign that could apply more broadly than the two specific participants.

We aimed to space the phases approximately two weeks apart for each participant. At the time of writing, we are partially through phase 2, having held the phase 2 session for Participant 1, and having scheduled the phase 2 session for Participant 2 for the immediate future.

2.2 Implementation

Choices for the design and implementation for each of the five interventions were grounded in a variety of approaches from literature and codesign feedback.

2.2.1 Camera Aiming.

The intervention for camera aiming, which happens in the foreground of our app, uses a combination of object recognition and edge detection. We use YOLOv8 [15], which is a one-shot object recognizers, to perform object recognition. Because of their one-shot technique, these models can detect objects very quickly, even within the app, for reduced latency in detection and feedback. If a user specifies what they are trying to center on, we first determine if that object is part of the COCO dataset [19]—the dataset on which YOLOv8 was trained. If it is, we search in the image specifically for

that object, and give feedback to bring it to the center of the frame. This does not require any API calls, so it can be done completely on device to make the feedback as real time as possible. If the object the user requests cannot be found in COCO, we alert the user to this fact, and proceed with the following method, which we also use in the case where the user does not specify any object to center the camera on. Here, we use YOLOv8 to take an initial pass on the image, and use the bounding boxes as well as confidence scores it returns to identify a salient object, and proceed to give feedback that brings it to the center of the frame if it is not already there. In addition to object detection via YOLO, we utilize edge detection to ensure the resulting framing does not cut off any objects. We utilize the results of object detection and the resulting bounding boxes of salient objects relative to the frame as a whole in tandem with the outcomes of edge detection. In doing so, we continue to offer the user centering/camera aiming feedback to improve the shot until a salient or requested object is centered and nothing is cut off. Once these criteria have been reached, if the intervention was called using Siri from a different app, the user is returned there. If the intervention was launched from our app directly, no such transition occurs, but the user is free to navigate to the camera app of their choice to make use of the centered frame. Though this technique diverges from much of the existing camera aiming literature for blind photography [4, 29, 31], which tends to gravitate towards methods like Lucas Kanade Optical Flow, we use this method instead to take advantage of more modern techniques that have emerged since these foundational papers' publication, and to more easily incorporate a user's direct intention in what gets centered. This intervention is still in iterative development, and we may adjust some aspects of this implementation in a final product.

2.2.2 *Check Photo Quality.*

The implementation of the intervention to check photo quality utilizes the model described in Mandal et al.'s 2023 work, *Helping Visually Impaired People Take Better Quality Pictures* [21]. This model, which is trained on photos taken by BVI individuals sourced from VizWiz [6] and Orbit [23] data sets, identifies the primary distortions present in an image among options of blurry, shaky, bright, dark, grainy, and none, and assigns each a value between 0 and 1 where a higher value denotes a greater contribution to the image's quality or lack thereof. We pass a user provided image to this model alongside a threshold for minimum significance of none for an image to qualify as high quality. We then determine which distortion is most prevalent within the image. If this distortion is not none, we provide the user with a corresponding description of the dominant quality issue in the image. If the top ranking distortion is none, we check if its impact score is at or above the threshold; if it is, we notify the user that the image they have provided is good quality, and if it is not, we select the second-highest ranked distortion and provide the user with a message that corresponds to this second-ranked distortion. This intervention is still in iterative development, and we may adjust some aspects of this implementation in a final product.

2.2.3 *Surface Unknown-Unknowns.*

The surfacing unknown intent aims to surface relevant information the user may not be aware of when they request information about their visual context. One particularly dangerous unknown unknown that frequently appears in these visual inquiries is duplicates of an object. One codesign participant resonated with how this could be potentially problematic, in reaction to a proposed example in which someone may want to know information about a sign for a bus route, but there were multiple signs, which would confound the output and make it difficult to determine the correct bus. To address this concern, when a user surfaces an intent targeted towards situational context (cross-checking or debugging) we use a combination of YOLOv8 [15] and YOLOworld [9] detection to determine if there are relevant duplicates a user is unaware of. To do this, we use GPT-4o to identify object classes that may be relevant to a user's inquiry. If a class is present in the Common Objects In Context (COCO) [19] dataset, we try detect it using YOLOv8 with a 75% confidence threshold. If the class is not part of COCO, we instantiate it with YOLOworld, and try to detect it with a confidence level of 5%. This confidence level is lower because YOLOworld, as an open vocabulary model, operates with much lower confidence thresholds even when detection is accurate since it is not trained on hyper-specific items but rather a broad offline vocabulary [30]. Both detection models are run on the image, and their outcomes are compiled into a single dictionary with classes found as keys and the quantity of said items as values. This is then compared to the generated classes of interest, and any classes of interest that were found multiple times are synthesized into a sentence describing the multiples found and their quantities. Finally, this sentence is checked against the user's original request for information to ensure the user was not expecting the duplicates that were found. If the duplicates were indeed unexpected, this information is presented to the user immediately after the response to whatever information they requested, be that the result of cross-check or debug.

2.2.4 *Cross-Check.*

The approach for the cross-check implementation was largely inspired by Huh et al.'s 2023 work, *GenAssist* [14]. This approach takes an expectation for what some visual content should be and an image and generates verification questions to check for correctness of each component of the information to be verified. In the original paper, where the goal is to help user's select which AI-generated image best meets their needs, these questions can then each be answered about each generated image by a visual-question answering model, and the differences between each image can then be summarized for the user to select which to use. In our case, since there is only one image, and the aim is to verify the statement's correctness, we instead employ a voting mechanism where multiple LLMs (GPT 4o and Gemini 1.5-flash) answer the verification questions. Because while LLMs are powerful, they also have a propensity to hallucinate, we also determine which, if any, verification questions can be answered by optical character recognition (OCR)—generally questions about text—or by an object recognizer—generally questions about the existence of some item. For questions that can be answered by OCR, an additional vote is cast based on the presence of the relevant text in the output of OCR provided by Google Cloud Vision. For questions that can be answered by object recognition, we combine the results of YOLOv8 [15] and YOLOworld [9] detection models, and cast a vote based on whether the specified object can be identified. The results of the vote for each verification questions are

combined to generate a comprehensive response about the accuracy of a user's expectation of an image.

2.2.5 *Debug.*

Though there has been little academic inquiry to date around specific mechanisms for BVI users to ascertain why a model performed in the way it performed, as many dominant explainable AI techniques are geared towards sighted users and are visual in nature [16, 20, 25], recent work by Alharbi et al. [5] documents BVI users' desire to for avenues to contest AI output and ascertain model reasoning to be built into visual assistance technologies. With this desire in mind, our intial phase of codesign revealed that a common strategy for debugging output is to pass the response of an AI along with the visual context that produced it to tools with vision capabilities (such as ChatGPT or Gemini) and ask why it produced that answer instead of the answer they expected to recieve. This conversation also revealed that if this process is undertaken in the original application—for example, through BeMyAI's chat feature—the outcome is often for the model to second guess itself even if the initial response was accurate, essentially instantiating hallucination. To replicate this technique, without the complications of having the original model overcorrect, we send the model output, the user's expectation, and the user's visual context to the server, and ask Gemini to summarize why the model might have responded in the way it responded. This reduces the labor associated with the debugging practice already in practice by many blind users, and separating the response from the specific system that produced it aims to reduce the likelihood of model overcorrection due to conversational history.

3 Preliminary Results

3.1 Surfacing Feedback

Prior to codesign, we envisioned that interventions that run automatically background would have the potential to more robustly capture visual contexts and be minimally disruptive to user workflows—both valuable for capturing information users may not know they need, such as unknown unknowns. However, in phase one codesign, we found that users perceive automatic, unannounced captures to be akin to 'looking over their shoulder', and that interventions that used this strategy would be seen more like spies and less like helpers. Users preferred intentional triggers for interventions for a couple of reasons. Firstly, intentional intervention triggers provided agency, allowing the user to define when they may want additional information and assistance and when they do not. Secondly, intentional triggers were preferable for privacy reasons, as they would not run the risk of automatic captures failing to stop when they were supposed to, which could lead to the accidental, unnecessary collection of personal information or private activity. To address concerns around awareness of context capturing, we implemented a sound alert whenever our app or a related shortcut takes a screenshot. To honor the preference for non-automatic intervention while still not requiring the user to somehow know what they are missing to intentionally ask to surface unknown unknowns, we integrate this intervention into other information finding interventions like cross-check and debug. This approach to activating the surface unknown-unknowns intervention balances the need for some form of automatic trigger, since the concept hinges on the user not recognizing the need for additional information beyond what has been asked for or provided, with a protection for user privacy that comes from the awareness that screen context is being captured.

3.2 Cross-System Accessibility

We chose to utilize Shortcuts due to the interface's potential for facilitating not just cross-tool integration, but also collection of user context and intention in calling interventions by leveraging existing operating system capabilities. However throughout testing and codesign, we have discovered several accessibility issues with the Shortcuts application, particularly pertaining to Voiceover [26], Apple's native screen reader. Firstly, the pop-up and input fields produced by Shortcuts are not Voiceover accessible—this includes permissions the Shortcut needs to be granted to run the first time it is called, as well as fields to provide input that is relevant to understanding a user's intention in calling a Shortcut. Secondly, with Voiceover on, it is not possible to add an action to Automations, which are Shortcuts configured to run automatically on a certain trigger, such as opening a specific app. To circumvent these issues, we plan to utilize Shortcuts to capture visual context through screenshots, but collect user-provided info either within our app or as part of the initial voice command that precedes the Shortcut running. This technique, however, does not resolve all Voiceover accessibility issues, because it fails to circumvent the permissions issue that exists the first time a shortcut is run. For this issue, we are actively investigating ways to pre-approve Shortcut actions from the general setting menu. To address the issue with adding actions to Automations, we plan to develop alternative surfacing methods for anything that would have otherwise required an Automation—such as running surface unknowns alongside cross check or debug instead of whenever a certain app opens.

4 Future/Ongoing Work

This work is ongoing. In the immediate future, we plan to complete the three proposed stages of codesign. Afterwards, we hope to validate the codesign findings through a lab-based user study. Ultimately, we hope to also evaluate the framework and intervention's ability to customize to organic user workflows through a deployment study, though this will not be feasible until the Shortcut Voiceover bugs are resolved on apple's side, given they pose difficulties for setup and recording which app called an intervention in order to return to it.

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